Certifiably Safe Manipulation of Deformable Linear Objects via Joint Shape and Tension Prediction

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IEEE INTERNATIONAL CONFERENCE ON ROBOTICS AND AUTOMATION

ATLANTA 2025

Overview

We propose a certifiably safe framework for manipulating deformable linear objects (DLOs) in contact-rich environments. The key contributions of our work include:

- A novel learning-based predictive model that jointly estimates future DLO shape and tension over extended time horizons, even under complex contact dynamics
- A trajectory optimization framework that incorporates these predictions to enforce certifiable safety constraints on tension and collision
- Our framework enables reliable DLO manipulation without overstretching or collisions and outperforms existing methods in a simulated wire harness assembly task.

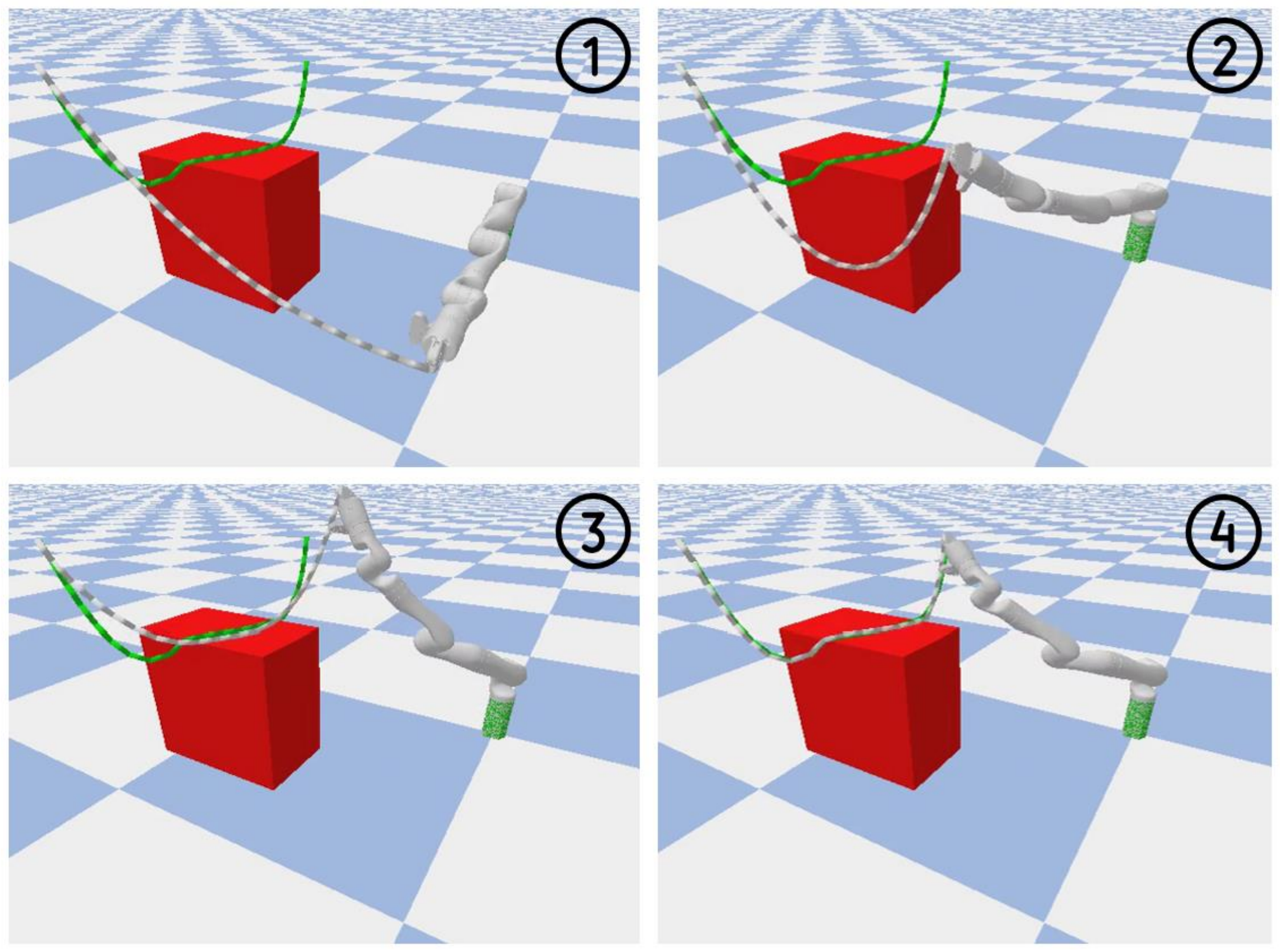

Figure 1. At the core of our method is a predictive model that simultaneously estimates the future shape and tension of the DLO. The figure illustrates a robot arm manipulating a DLO (white) toward a goal configuration (green) in the presence of an obstacle (red). The manipulation is executed without causing collisions or overstretching, despite contact with the environment.

Methodology

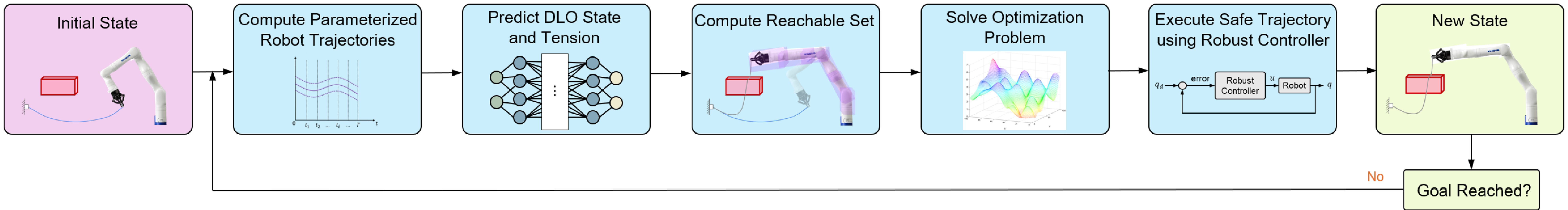

Figure 2. Overview of the proposed framework. Starting from an initial state, the robot generates a set of parameterized trajectories and uses a learned predictive model to estimate the resulting DLO shape and tension. These predictions are incorporated into a reachable set computation, enabling the identification of safe trajectories under contact-rich conditions. An optimization problem is then solved to select the best feasible trajectory, which is executed using a robust controller. The loop runs in a receding horizon fashion until the DLO reaches the desired goal configuration without collisions or overstretching.

Figure 3. Overview of the discretized configuration of DLO.

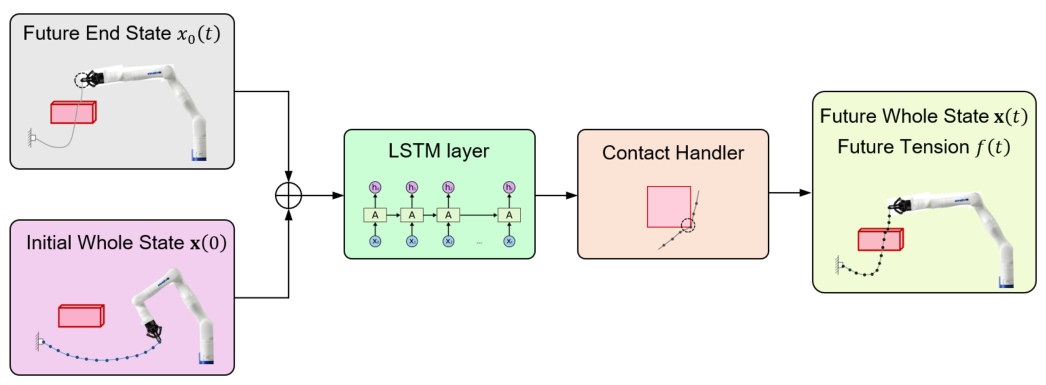

Figure 4. Architecture of the DLO shape and tension prediction model.

Safety Constraints

The planned trajectory must respect the robot's joint position, velocity, and torque limits while ensuring collision avoidance. To guarantee safe DLO manipulation, we enforce a maximum allowable tension threshold f_{lim}, requiring all predicted tensions along the trajectory to remain below this limit.

Cost Function

We define the accumulated distance between the DLO's state in the trajectory and the desired goal state as the cost function of the trajectory optimization problem

$$
\begin{aligned}
&\min_{k \in K} \quad \texttt{cost}(k) \\
&\mathbf{q_j}(\mathbf{T_i};k) \subseteq [q^-_{j,\text{lim}}, q^+_{j,\text{lim}}] && \forall i \in N_t, j \in N_q \\
&\dot{\mathbf{q}}_\mathbf{j}(\mathbf{T_i};k) \subseteq [\dot{q}^-_{j,\text{lim}}, \dot{q}^+_{j,\text{lim}}] && \forall i \in N_t, j \in N_q \\
&\mathbf{u}(\mathbf{q_A}(\mathbf{T_i};k), \mathbf{f}(\mathbf{T_i};\mathbf{K})) \subseteq [u^-_{j,\text{lim}}, u^+_{j,\text{lim}}] && \forall i \in N_t, j \in N_q \\
&\mathbf{FO_j}(\mathbf{q}(\mathbf{T_i};k)) \bigcap O = \emptyset && \forall i \in N_t, j \in N_q \\
&\|\mathbf{f}(\mathbf{T_i};\mathbf{K})\| < f_{\text{lim}} && \forall i \in N_t.
\end{aligned}
$$

Results

We evaluate the proposed method in a simulated warehouse environment. The task requires the robot to manipulate the wire harness safely along a planned trajectory from an initial state to a goal state. Initially, the wire harness is held in free space without contacting the engine. In the goal state, the middle section of the wire harness must be placed securely onto the engine.

Baselines

(1) *Learning Where to Trust*, which uses a classifier to identify robot motions that might lead to trapping. However, it lacks certifiable safety guarantees for both the robot and the DLO.

(2) *ARMOUR*, a certifiably safe planner that ensures robot safety but does not account for DLO-specific risks such as overstretching.

Table 1. Simulation results under random environment setups.

Method	Goal Reached	Failed w/o Violation	Robot Collision	DLO Overextension
Learning Where to Trust	38	30	29	3
ARMOUR	54	20	0	26
Ours	76	24	0	0